

Yousif Ahmed

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Professional Summary

Software Engineering graduate specializing in AI-powered backend systems and full-stack development, experienced designing Retrieval-Augmented (RAG) pipelines, NLP and computer vision applications, and scalable financial platforms. Passionate in building reliable software that combines AI with real-world business solutions.

Experience

Internship At Quenet Communication Systems Jun 2025 – Jul 2025

- Designed and deployed a real-time full-stack dashboard integrating Qmatic APIs to visualize live branch, queue, and service-point KPIs for operational decision-making.
- Developed Node.js/Express data pipelines to aggregate, normalize, and analyze real-time operational data, generating metrics and alerts for queue performance.
- Technologies: React, TypeScript, Node.js, Express, Tailwind CSS

Education

Cyprus International University, BS in Software Engineering. Oct 2022 – Jun 2026

- CGPA: 3.22/4.0

Golden Visa Recipient: Awarded by UAE Government for High School Academic Excellence. Sep 2022

Projects

InVox AI-Driven Invoice Intelligence System (Capstone) Feb 2026 – present

- Built a 9-stage OCR + LLM invoice ingestion pipeline that reduced processing time by 8.0×, automating data extraction, compliance validation, duplicate detection, vector embedding generation, and invoice lifecycle automation.
- Engineered a hybrid RAG pipeline combining pgvector retrieval, query rewriting, context compression, cross-encoder reranking, and intent classification to route SQL, semantic, and hybrid queries while grounding financial responses in database results.
- Designed a provider-agnostic AI architecture supporting runtime switching between Gemini, OpenAI, Grok, Ollama, and rerankers through environment configuration with zero application code changes.
- Implemented a 5-rule compliance engine with automatic flag resolution, validated on 1000+ synthetic invoices spanning 10 vendors, 8 clients, and 13 months of financial activity.
- Technologies: Python, FastAPI, React, PostgreSQL, Supabase, pgvector, Gemini, Ollama, Tailwind CSS.

Mental Health Text Classifier. NLP / Deep Learning May 2026

- Fine-tuned DistilBERT on 42,380 labeled mental health statements across 7 classes (83% validation accuracy, macro F1 0.81), Integrated FAISS semantic retrieval with classifier
- Evaluated model performance using confusion matrices and per-class precision, recall, and F1, identifying class imbalance and limitations
- Technologies: Python, HuggingFace, Transformers, PyTorch, FAISS, Streamlit.

Football Match Analysis. Computer Vision (Independent) Feb 2025

- YOLOv8 object detection pipeline for player and ball tracking across match footage.
- Includes team assignment via K-means jersey color clustering, optical flow camera compensation, perspective transformation to pitch coordinates, and per-player speed and distance calculation.
- Technologies: Python, OpenCV, PyTorch. Ultralytics. Roboflow

Key Skills

- **Languages:** Python, JavaScript/TypeScript, C++
- **AI / Machine Learning:** Retrieval-Augmented Generation (RAG), Transformer Models, Tool Orchestration, Vector Databases, Intent Classification, Prompt Engineering, NLP, Computer Vision
- **Frameworks & Libraries:** FastAPI, React, Node.js, PyTorch, Hugging Face Transformers, OpenCV, Ultralytics
- **Databases & Vector Stores:** PostgreSQL, pgvector, MySQL, FAISS, Supabase
- **Languages:** Arabic (Native), English (Fluent)